

ANALYSIS REPORT

EV Battery-Swap Network: Service Reliability Diagnostics & Rider Churn Risk

Diagnosing no-battery failure drivers, quantifying network economics, and predicting rider churn to prioritise retention and reliability investment

Data period: January 2024 – June 2025 (18 months)

Coverage: 6 cities • 152 stations • 19,950 active riders • 3.88M swap attempts

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1. Problem Understanding

The business operates a **battery-swap charging network for electric two-wheelers and three-wheelers** across six Indian cities: Bengaluru, Delhi NCR, Hyderabad, Jaipur, Mumbai and Pune. Riders visit a station, exchange a depleted battery for a charged one, and pay per swap. Over the 18-month observation window, the network recorded **3.88 million swap attempts** across **152 stations**, serving **19,950 active riders** who are either independent riders or affiliated with one of **12 fleet partners**.

Two critical business problems prompted this analysis:

- **Service reliability** — in ~3.5% of visits, a rider finds **no charged battery available** (a “no-battery failure”). This is the single largest controllable driver of a poor rider experience, and it is not evenly distributed across cities, stations or seasons.
- **Rider retention** — roughly one in three riders **churns** (goes inactive) within the observation window, resulting in the loss of recurring revenue. Churned riders account for different amounts of revenue, so the business needs to know who is at risk and how much revenue that represents.

Business questions this report answers

- What operational and environmental factors drive no-battery failures, and can the failure rate be predicted well enough to plan ahead?
- Which stations, cities and station “expansion waves” (rollout cohorts) underperform, and why?
- What are the network's core unit economics (revenue, energy cost, contribution margin)?
- What actually predicts rider churn — is it service quality, usage behaviour, or rider segment?
- Which riders account for the greatest revenue at risk, and how should retention efforts be prioritised?

Data sources

Seven linked datasets were used and cross-validated for referential integrity (zero orphaned rider, station or battery references across all joins):

Dataset	Rows	What it captures
stations.csv	152	Station metadata: city, location/host type, expansion wave, charger generation, rent and tariff economics
station_hourly_status.csv	1,487,712	Hourly telemetry per station: charged packs, quarantined packs, chargers online, temperature, outage minutes
swap_events.csv	3,877,013	Every swap attempt: outcome, pricing, battery state of charge and state of health, queue wait, payment
riders.csv	20,000	Rider profile: vehicle class, plan type, fleet affiliation, sign-up channel, home city
batteries.csv	6,500	Battery pack inventory: supplier, cost, rated capacity, state of health
fleet_partners.csv	12	B2B partner contract terms and discount structures
support_tickets.csv	44,000	Rider support interactions: category, priority, resolution time, CSAT

2. Analytical Approach

The analysis was organised into two linked tracks that share the same station, rider and financial context:

Track A — Service Reliability & Network Economics

- Merged 18 months of hourly station telemetry with the swap-attempt log and station metadata, then aggregated the data by city and month.
- Ran correlation analysis between the no-battery failure rate and nine candidate drivers (temperature, grid outages, heat-alert days, rainfall, flood days, packs quarantined/charging, chargers online, low-battery hours).
- Fitted a multiple linear regression model to predict the failure rate from the strongest drivers, then examined residuals to find where and why the model overpredicts or underpredicts.
- Compared failure rates across station “expansion waves” (Launch vs. two later rollout cohorts) to test whether newer stations perform better.
- Built a station-level anomaly score (blending failure rate and low-battery hours) to flag the highest-priority stations.
- Reconstructed monthly network financials — revenue, energy cost and contribution margin — from completed-swap pricing data.

Track B — Rider Churn Prediction

- Defined “churn” as no swap activity in the final 60 days of the observation window, for riders with enough history to be evaluated fairly.
- Built a rider-level feature set: usage volume, service experience (failure rate, queue wait), support-ticket history, revenue, and segment (fleet vs. independent, plan type, vehicle class).
- Examined univariate churn-rate breakdowns for every segment and behavioural grouping to identify which factors distinguish churned riders from retained riders.
- Assembled a 24-column, 19,950-row modelling dataset and trained three classifiers — Logistic Regression, Random Forest and Gradient Boosting — on an 80/20 train-test split with one-hot-encoded categorical variables.
- Evaluated models on accuracy, precision, recall, F1 and ROC-AUC, then extracted feature importances to identify the main predictors of churn.
- Converted model output into a churn-probability score, grouped riders into Low, Medium and High risk bands, and quantified revenue exposure by risk band and by rider segment.

Tools & techniques

Python (pandas, numpy) for data preparation; Pearson correlation and ordinary least-squares regression for driver analysis; scikit-learn Logistic Regression, Random Forest and Gradient Boosting classifiers for churn prediction; precision/recall/F1/ROC-AUC and confusion matrices for model evaluation.

3. Key Insights

3.1 What drives no-battery failures

The overall network no-battery failure rate is **3.47%** of all swap attempts, but this overall figure masks substantial, systematic variation. Correlating the city-month failure rate with operational and environmental factors provides a clear ranking of drivers:

Driver	Correlation	Relationship
Average maximum temperature	0.65	Strong positive
Grid outage hours	0.59	Moderate positive
Heat-alert days	0.57	Moderate positive
Average packs quarantined	0.55	Moderate positive
Low-battery hour %	0.43	Moderate positive
Flood days / total rainfall	−0.05 to −0.06	Negligible
Packs charging / chargers online	0.02–0.02	Negligible

A **multiple linear regression** built on these drivers explains a substantial share of the variation in monthly city-level failure rates: **$R^2 = 0.663$** , MAE = 1.02 percentage points, RMSE = 1.49 percentage points. Model error varies by city. It is lowest in Mumbai and Bengaluru (average error < 0.9 percentage points) and highest in Delhi NCR and Jaipur (average error ~1.3–1.4 percentage points), which means that a single national model performs less well in the hottest cities with the most variable failure rates. City-specific tuning would help in these locations.

Failure rates vary substantially by city:

City	Failure rate	Attempts	Failures
Jaipur	4.78%	487,634	23,303
Delhi NCR	4.41%	768,913	33,892
Hyderabad	3.97%	711,930	28,274
Pune	2.80%	552,910	15,480
Bengaluru	2.50%	825,327	20,602
Mumbai	2.42%	530,299	12,838

The worst city-month was **Jaipur in May 2024, when the failure rate reached 12.94%**, nearly 4x the network average. The monthly trend shows a clear **summer spike** each April–June, driven by heat and concurrent grid stress rather than a one-off event.

Newer stations are structurally more reliable. Grouping stations by rollout cohort (“expansion wave”) shows the original Launch-wave stations failing at **3.55%**, versus **2.77%** for the first later wave and **2.72%** for the most recent wave — roughly a full percentage point of improvement, consistent with improvements in hardware generation, siting or battery-buffer sizing based on experience after launch.

The top 15 highest-failure and highest-anomaly stations are predominantly Launch-wave stations in **Jaipur and Delhi NCR with Gen1 8-slot hardware** with unusually high “low-battery hours” (over 200 hours/month at the worst sites). The same sites appear on both the failure-rate and anomaly-score rankings, confirming that these locations have persistent, fixable problems rather than random fluctuations.

3.2 Network economics

Metric	Value
Total revenue (18 months)	₹23.31 Cr
Total energy cost	₹7.11 Cr
Total contribution margin	₹16.20 Cr (69.5% of revenue)
Average contribution per completed swap	₹43.25
Overall failure rate	3.47%
Overall abandonment rate	1.77%

Revenue and contribution both grew roughly 3x from the first month to the last, and contribution per swap has also risen (from ~₹38 to ~₹48). The network's variable-cost economics are healthy and improving, which directly affects how much investment in reliability and retention it can justify.

3.3 What actually drives rider churn

Across the 19,950 riders with enough history to evaluate, **33.88% churned** (6,760 riders) within the observation window. Breaking down churn by segment shows almost **no meaningful difference** by fleet affiliation (Fleet Partner 33.95% vs. Independent 33.78%), plan type (33.3–34.0% across all three plans), or vehicle class (2W 33.55% vs. 3W 35.87%). None of these main commercial segments explains churn on its own.

Usage intensity provides the strongest and most counterintuitive signal, ahead of service quality:

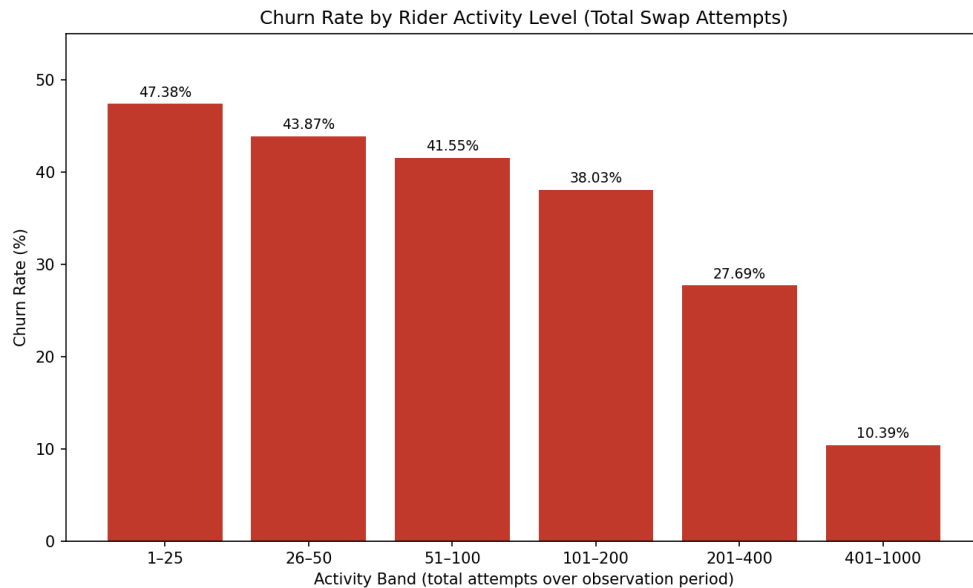


Figure 10. Churn rate by total swap attempts over the observation period.

The churn rate is 47.4% for riders with only 1–25 total attempts and falls steadily to **10.4%** for riders with 401–1,000 attempts. By contrast, churn is **inversely** related to the failure rate experienced: riders with a 0–2% personal failure rate churn **more** (49.3%) than riders with a 15%+ failure rate (28.8%), and churned riders logged fewer support tickets on average (1.33 vs. 2.65) than retained riders. This pattern is consistent with a **tenure/cohort effect**: riders who signed up recently have had less time to accumulate attempts, tickets and failures before the churn cutoff, and can therefore appear to be “churners” in a fixed-window definition even though nothing about their experience necessarily drove them away. This is explicitly identified as a modelling caveat in Section 6.

However, there is still a signal related to service failures: among riders who **did** raise a support ticket, the ticket **category** is informative. **Billing disputes** have the highest association with churn of any category (**31.7%**), ahead of long-queue complaints (23.0%) and no-battery-available reports (22.4%), and well above generic app issues (16.9%) or “other” (16.0%). Billing friction appears to be a specific, fixable issue affecting retention, even though the aggregate failure rate does not show the same relationship.

3.4 Predicting churn: model results

Three classifiers were trained on the same 38 processed features (19 numeric features, with the remainder consisting of one-hot-encoded categorical variables):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	64.3%	48.1%	68.9%	56.7%	70.6%
Random Forest	75.6%	61.9%	73.3%	67.1%	83.5%
Gradient Boosting	81.5%	91.8%	49.9%	64.6%	86.6%

Gradient Boosting has the highest overall accuracy and ROC-AUC, but its recall on the churn class is only **49.9%**, so it misses roughly half of all riders who actually churn. **Random Forest** trades some precision for a much higher **73.3% recall**, catching most at-risk riders while remaining reasonably accurate. This makes it the better choice for an early-warning retention system, where missing a rider who will churn is more costly than one additional false alarm.

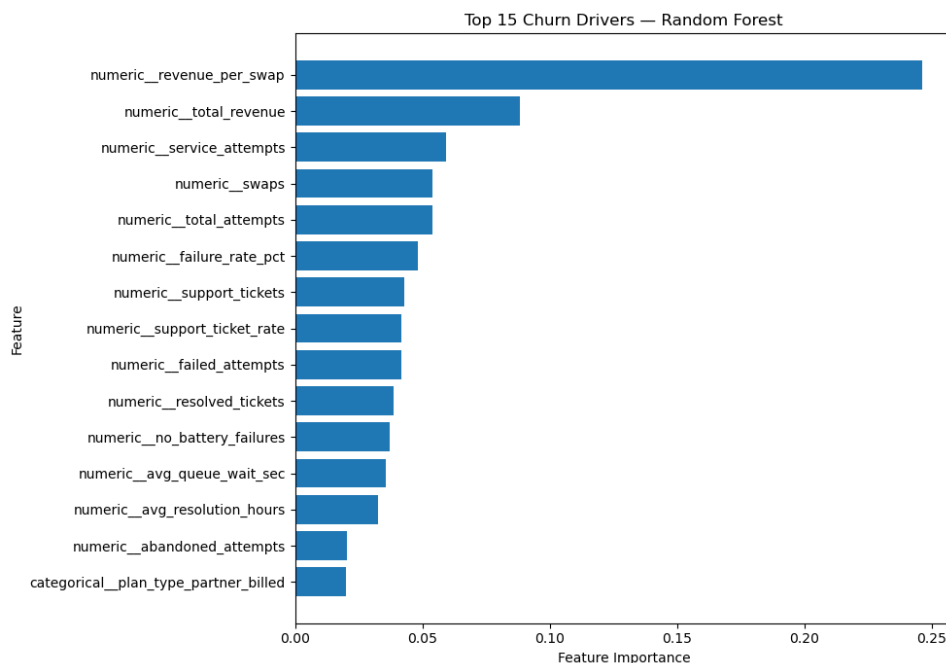


Figure 9. Top churn drivers by Random Forest feature importance.

Feature importance confirms the pattern associated with usage intensity: **revenue per swap (24.6%)** and **total revenue (8.8%)** are by far the strongest predictors, followed by service attempts, total swaps and total attempts. **Failure rate (4.8%)** and support-ticket metrics rank considerably lower. In effect, the model mainly learns how much riders use the service, with less emphasis on the quality of their service experience.

3.5 Where the revenue risk is concentrated

When the 3,990-rider test set is scored, **656 riders (16.44%)** fall into the High Risk churn band, **1,072 (26.9%)** into Medium Risk, and **2,262 (56.7%)** into Low Risk. High-risk riders account for **₹47.79 lakh** of test-set revenue — **10.41%** of the total — at an average of ₹7,285 per rider.

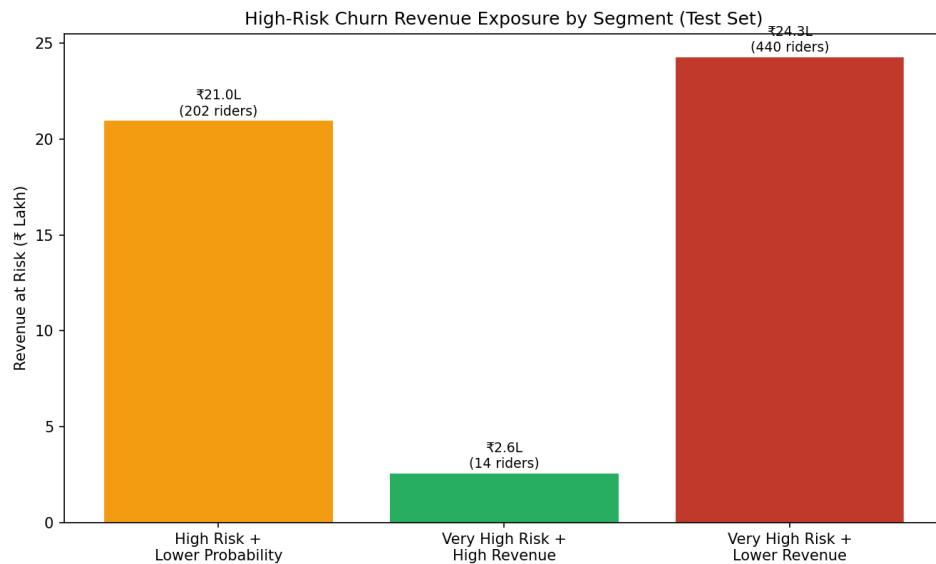


Figure 11. Revenue at risk split by churn-probability and revenue segment (test set).

Revenue at risk is also concentrated: the top 66 high-risk riders (90th percentile by revenue) represent **27%** of all at-risk revenue, and **2W, partner-billed, fleet-partner riders make up 54.7%** of the at-risk revenue pool — by far the largest single exposure segment. A more detailed breakdown of risk shows that a small group of **14 “Very High Risk + High Revenue” riders** (average revenue of ₹18,244 each) warrants individual, personalised retention support, while the much larger **440-rider “Very High Risk + Lower Revenue” group** (50.8% of at-risk revenue) is better served by automated, lower-cost retention offers.

4. Supporting Visualizations

The charts below were produced directly from the underlying analysis and are referenced throughout Section 3; they are collected here for easy review.

Reliability & Operations

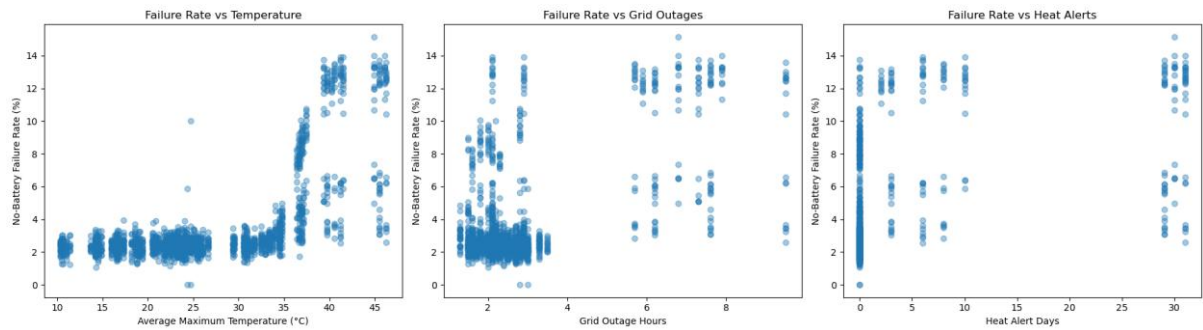


Figure 1. No-battery failure rate vs. temperature, grid outages, and heat-alert days (city-month level).

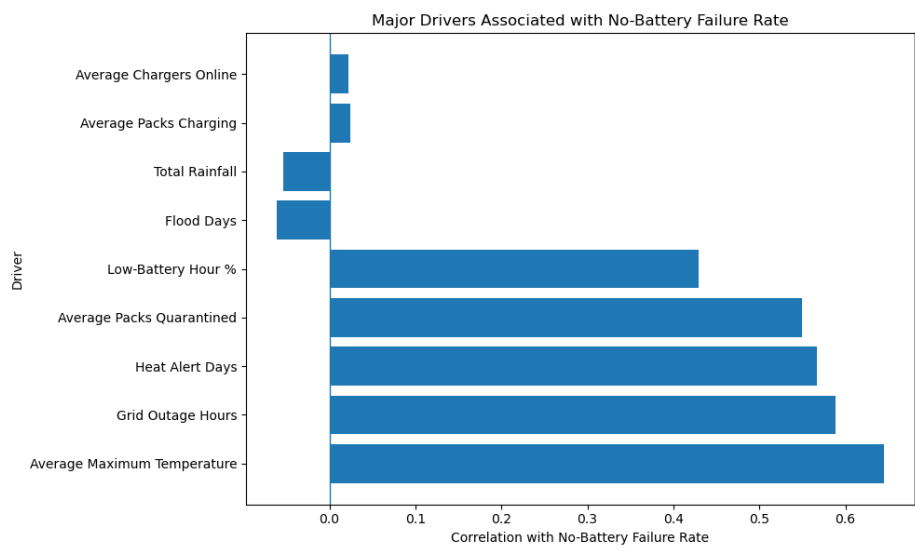


Figure 2. Correlation strength of each operational or environmental driver with failure rate.

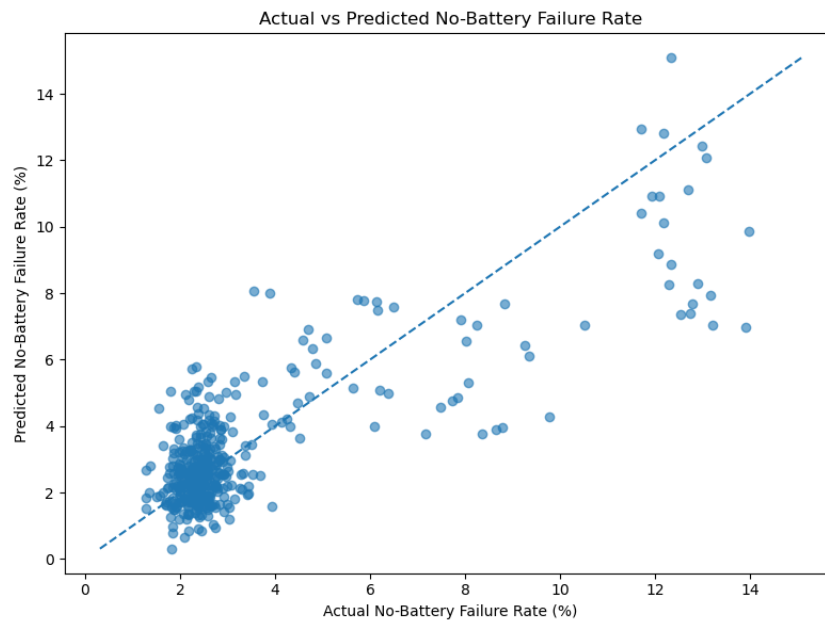


Figure 3. Actual vs. predicted failure rate from the regression model ($R^2 = 0.66$); points near the dashed line have accurate predictions.

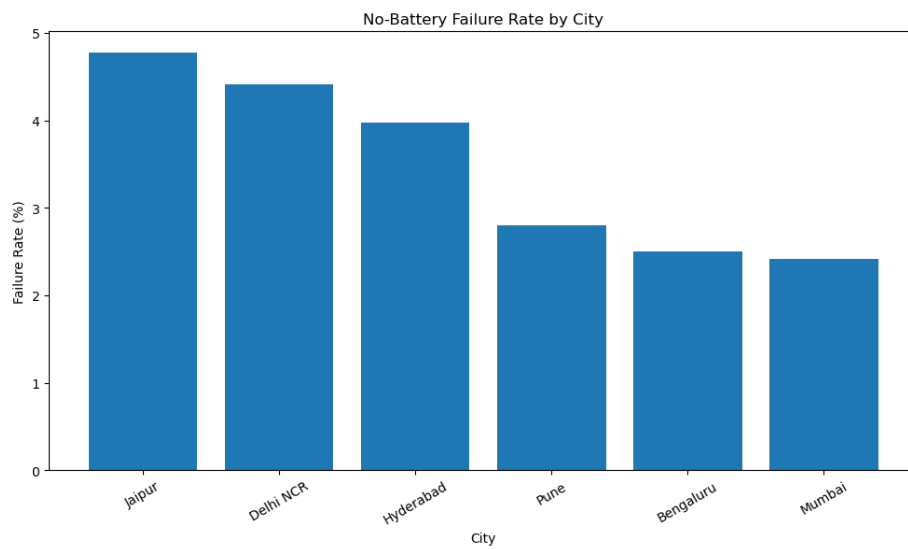


Figure 4. No-battery failure rate by city.

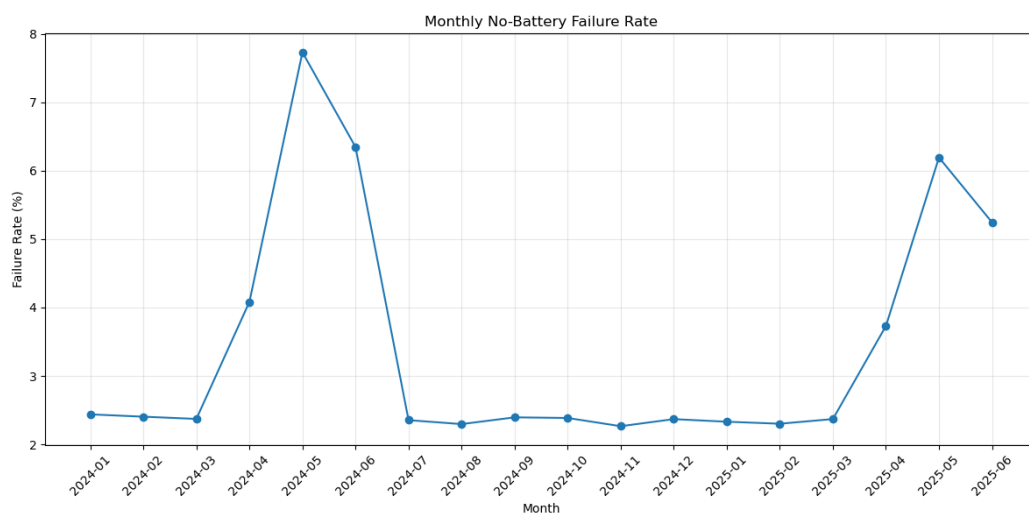


Figure 5. Monthly no-battery failure rate, Jan 2024 – Jun 2025 — note the repeating April–June spike.

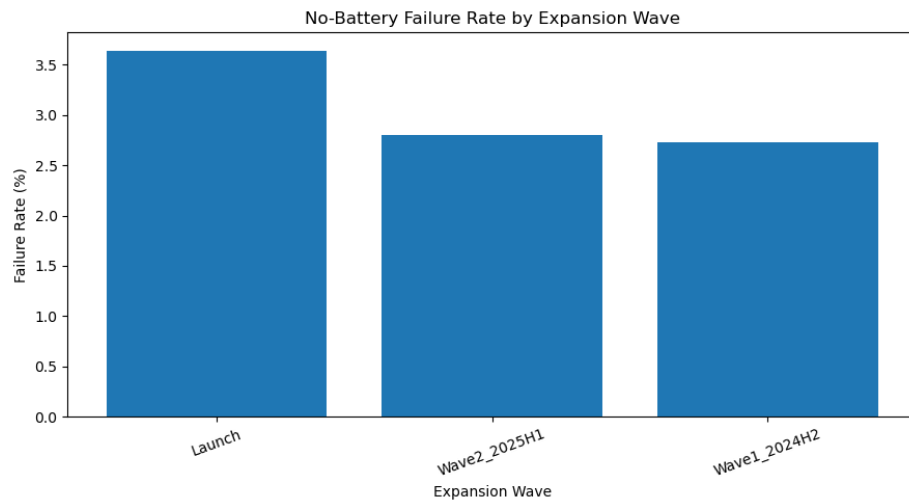


Figure 6. Failure rate by station expansion wave — later cohorts outperform the original Launch wave.

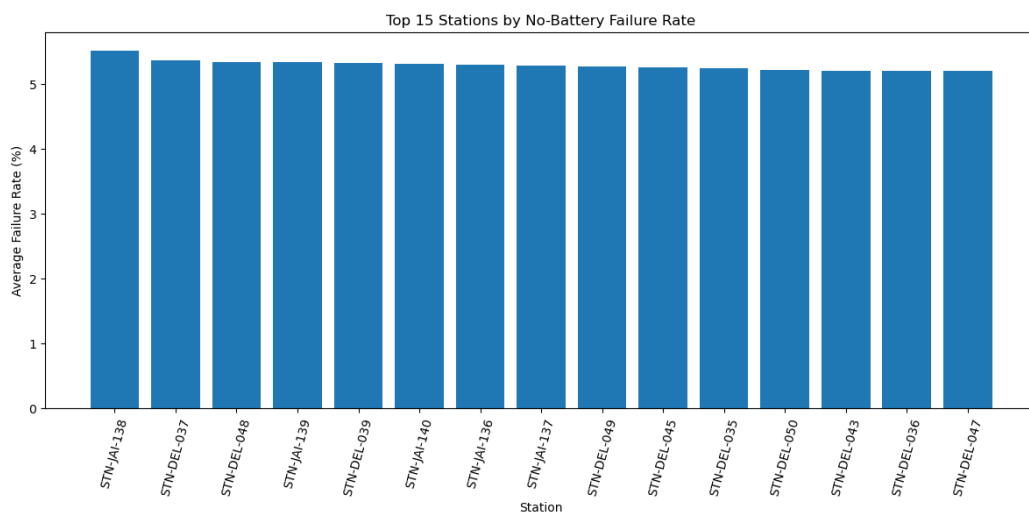


Figure 7. Top 15 stations by average no-battery failure rate.

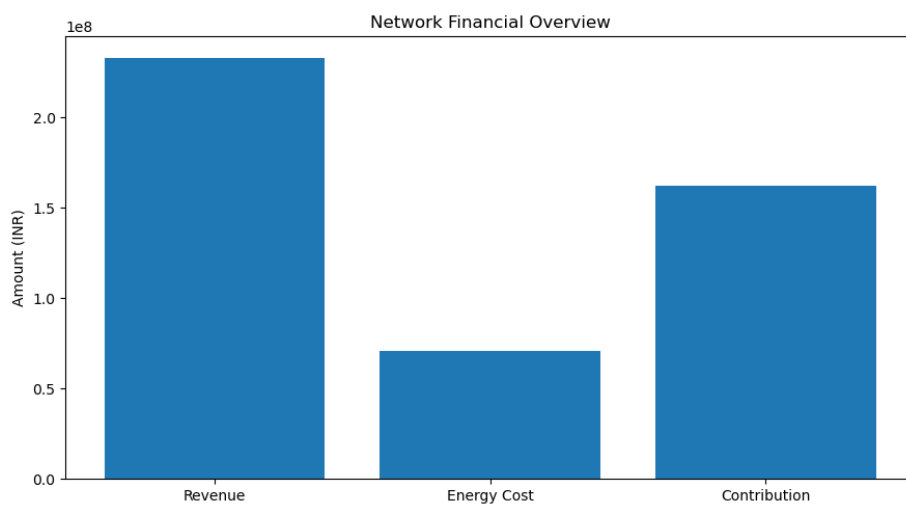


Figure 8. Network revenue, energy cost, and contribution margin (18-month total).

Rider Churn

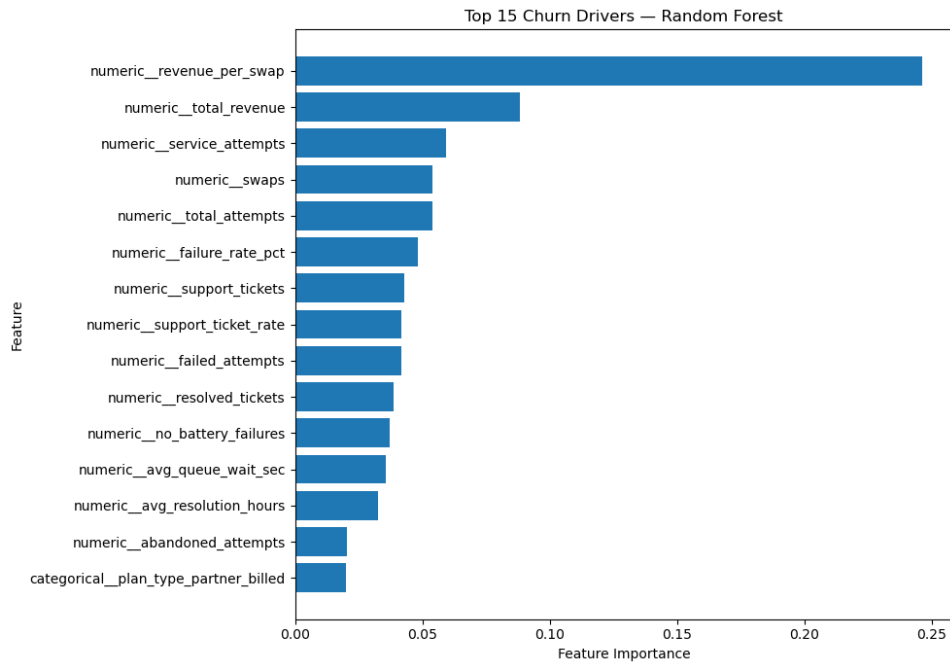


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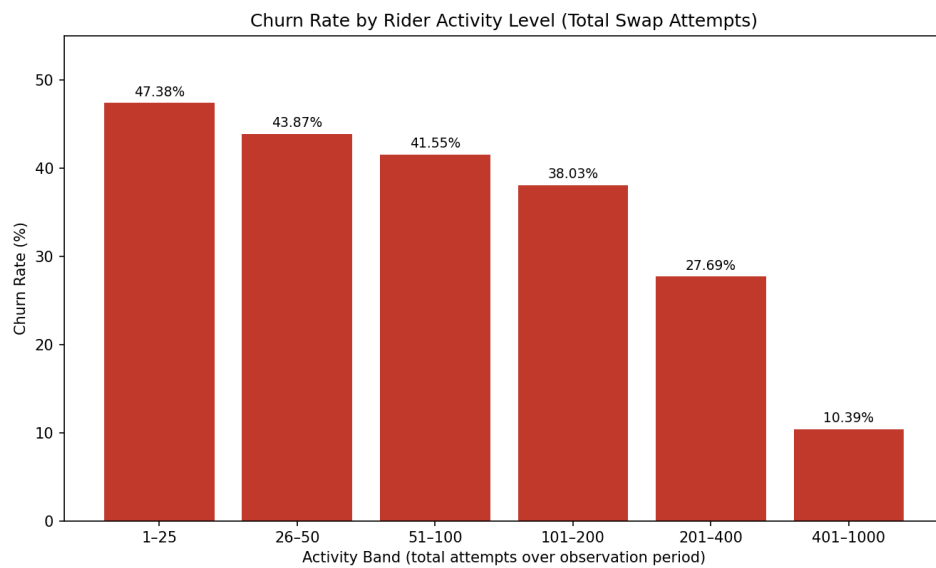


Figure 10. Churn rate by rider activity level (total swap attempts).

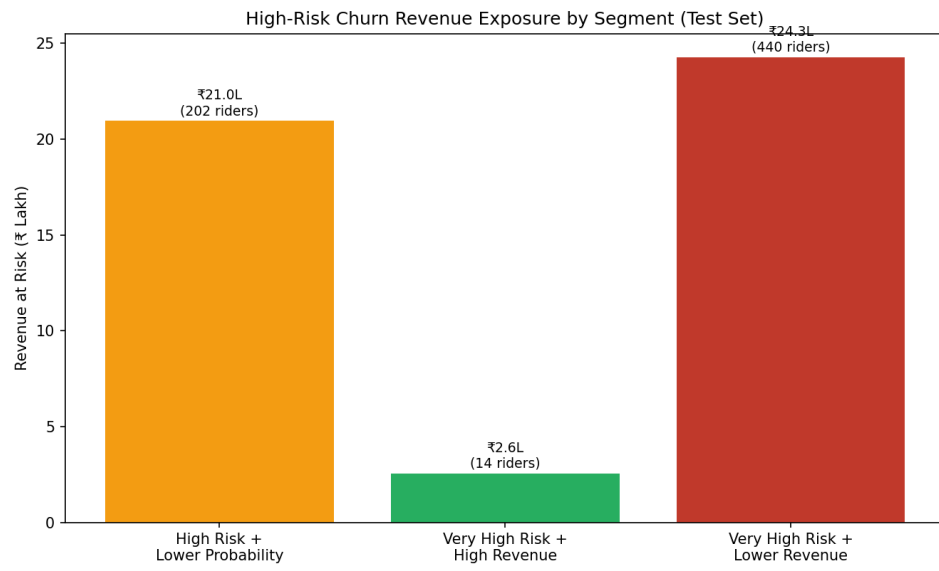


Figure 11. Revenue at risk by churn-risk and revenue segment (test set).

5. Business Findings

The analysis has the following implications for the business:

Heat is the single biggest, most predictable operational risk

The failure rate rises sharply above roughly 35°C rather than increasing gradually with temperature, and the network's two hottest, most heat-exposed cities (Jaipur, Delhi NCR) have both the highest average failure rates and the worst individual city-months. Because this pattern is seasonal and recurs every April–June, it calls for advance planning rather than a response to an unexpected event.

Grid instability compounds the heat problem

Grid outage hours correlate with failures almost as strongly as temperature itself (0.59 vs. 0.65), and the two tend to occur together. This points to insufficient battery buffers or backup power at exactly the stations and times when demand for charged batteries is highest.

The network has improved structurally — newer stations perform better

Post-launch expansion waves have failure rates roughly one percentage point lower than the original Launch cohort, and the worst-performing stations today are almost all old, Gen1, Launch-wave sites in Jaipur and Delhi NCR. This is encouraging operationally: the improvements in siting, buffer sizing and hardware are already known to work, but have not yet been applied to the legacy station fleet.

The business can afford to invest in fixing this

At a 69.5% contribution margin and roughly 3x revenue growth over 18 months, the network's unit economics are healthy and improving. The case for spending on station upgrades and retention programmes is strong because the underlying business can absorb the cost.

Churn is a scale/engagement problem, not primarily a quality problem

With roughly a third of riders churning and virtually no relationship to failure rate or fleet/plan segment, broad messaging focused on reducing failures to cut churn is unlikely to have much effect. Churn is concentrated among low-usage, likely newer riders. This appears to be more of an onboarding and habit-formation problem than a service-quality problem, and should be measured and addressed accordingly (see Section 6's note on churn-definition bias).

Billing friction is a real, fixable exception to that pattern

Even though overall failure rate does not predict churn well, riders who raise a billing dispute churn at the highest rate of any support-ticket category (31.7%). This is a specific, addressable process issue rather than a broad reliability one.

Retention effort has an 80/20 structure

A small number of riders account for a disproportionate share of at-risk revenue: the top 66 high-risk riders alone represent 27% of at-risk revenue, and over half of all at-risk revenue comes from 2W, partner-billed, fleet-partner riders. Spreading the retention budget evenly across all flagged riders would waste effort relative to targeting these groups.

Random Forest is the more useful churn model for this business, despite lower headline accuracy

Because a missed rider who churns (and the resulting loss of recurring revenue) is more costly than one extra unnecessary retention offer, Random Forest, which identifies the most riders who actually churn at 73.3% recall, is more valuable operationally than Gradient Boosting at 49.9% recall, even though Gradient Boosting achieves higher overall accuracy.

6. Actionable Recommendations

A. Network Operations & Reliability

- 1. Build up battery buffers before summer begins.** Increase charged-battery stock and the priority given to clearing quarantined batteries at Jaipur and Delhi NCR stations by an estimated 20–30% from March through June, using the observed April–June spike to trigger advance planning rather than reacting after failure rates climb.
- 2. Target backup power at the stations with the highest anomaly scores first.** The ~15 stations flagged as top failure/anomaly outliers (mostly Jaipur/Delhi NCR, Gen1, Launch-wave) are where outage hours and battery quarantine rates compound the problem. Prioritise UPS and backup-power and battery-buffer upgrades there before a network-wide rollout.
- 3. Retrofit legacy Launch-wave stations with the improvements used in later waves.** Since Wave1/Wave2 stations have failure rates ~0.8–1.0 percentage points lower, treat the highest-anomaly Launch-wave sites as priority candidates for the same hardware, siting or battery-allocation changes used in later cohorts, rather than commissioning only new stations.
- 4. Use the regression model as an input to monthly planning.** Feed forecast temperature and grid-outage data into the model each month to flag city-months likely to see a spike in failure rates, and use it to pre-position batteries and staff. At $R^2 = 0.66$ it is reliable enough for city-level planning, but not for single-station decisions. Pair it with local judgement in Delhi NCR and Jaipur specifically, where model error is highest.

B. Retention & Rider Experience

- 5. Prioritize early-tenure engagement over failure-reduction messaging.** Since churn is primarily associated with low usage and short tenure rather than failure exposure, invest in onboarding nudges and usage milestones (e.g., first 25/50/100 swaps) for new riders. Service-reliability fixes alone are unlikely to substantially change overall churn.
- 6. Fix billing-dispute handling as a standalone workstream.** Billing disputes have the highest association with churn of any ticket category (31.7%). Audit discount and tariff logic and shorten resolution times specifically for this category.
- 7. Deploy the Random Forest risk score as an operational early-warning system.** Score active riders monthly and direct the ~16% flagged “High Risk” to a retention workflow (targeted offers and priority support). The model identifies roughly three-quarters of eventual churners at acceptable precision.
- 8. Concentrate retention spending on the two highest-value at-risk groups rather than distributing it evenly across all 656 flagged riders.** Give the 14 “Very High Risk + High Revenue” riders (~₹18K average revenue) individual, personalised outreach; give the cohort of the top 66 high-risk riders by revenue (27% of at-risk revenue) targeted, proactive win-back offers before they lapse.
- 9. Work through fleet partners, not just individual riders, for the largest segment.** 2W, partner-billed, fleet-partner riders account for 55% of at-risk revenue. Pair rider-level retention offers with partner-level SLA and incentive conversations for this segment.

C. Measurement & Next Steps

- 10. Re-examine the churn definition before acting further on it.** The current label of 60-day inactivity is confounded with rider tenure: new sign-ups can appear to have churned before they have had time to build a usage history. A cohort-adjusted or survival-analysis approach would separate genuine attrition from natural ramp-up and substantially improve every subsequent insight into churn.
- 11. Extend the failure model with station-specific variables.** The largest prediction errors (up to ~7 percentage points) and city-level error differences suggest missing local factors. Staffing levels, footfall or nearby competitor density are logical next additions.